

Solubility and Colligative Properties

Answer key with worked solutions

A. Understanding

1. Definitions.

(a) A property that depends only on the **number** of dissolved particles, not their identity. (b) Moles of solute per **kilogram of solvent** (mol/kg). (c) The number of particles each formula unit gives in solution. (d) Dissolved particles lower the **vapour pressure** of the solvent (fewer solvent molecules escape from the surface), so a higher temperature is needed for the vapour pressure to reach atmospheric pressure.

2. Solubility.

(a) Sugar has many -OH groups and forms **hydrogen bonds** with water; hexane is non-polar and cannot attract it (like dissolves like). (b) **Gases are less soluble** in warm water, so there is less dissolved oxygen. (c) The drink is bottled under high CO₂ pressure; opening it **lowers the pressure**, so by Henry's law less CO₂ can stay dissolved and it forms bubbles.

B. Freezing and boiling points

3. Glucose.

$$(a) n = \frac{18.0 \text{ g}}{180.16 \text{ g/mol}} = 0.09991 \text{ mol}; m = \frac{0.09991 \text{ mol}}{0.100 \text{ kg}} = \mathbf{0.999 \text{ mol/kg}}$$

$$(b) \Delta T_f = 1 \times 1.86 \frac{^{\circ}\text{C kg}}{\text{mol}} \times 0.999 \frac{\text{mol}}{\text{kg}} = 1.86 ^{\circ}\text{C}, \text{ so it freezes at } \mathbf{-1.86 ^{\circ}\text{C}}$$

$$(c) \Delta T_b = 1 \times 0.512 \frac{^{\circ}\text{C kg}}{\text{mol}} \times 0.999 \frac{\text{mol}}{\text{kg}} = 0.512 ^{\circ}\text{C}, \text{ so it boils at } \mathbf{100.512 ^{\circ}\text{C}}$$

4. Order of freezing points.

Highest first: **glucose** (i = 1, 0.10 mol/kg of particles), **NaCl** (i = 2, 0.20 mol/kg), **CaCl₂** (i = 3, 0.30 mol/kg). More particles give a larger depression, so a lower freezing point.

5. Calcium chloride.

$$n = \frac{11.1 \text{ g}}{110.98 \text{ g/mol}} = 0.1000 \text{ mol}; m = \frac{0.1000 \text{ mol}}{0.250 \text{ kg}} = 0.4001 \text{ mol/kg}$$
$$\Delta T_f = 3 \times 1.86 \frac{^\circ\text{C kg}}{\text{mol}} \times 0.4001 \frac{\text{mol}}{\text{kg}} = 2.23 \text{ }^\circ\text{C}, \text{ so it freezes at } -2.23 \text{ }^\circ\text{C}$$

6. Molar mass from a boiling point.

$$m = \frac{\Delta T_b}{iK_b} = \frac{0.256 \text{ }^\circ\text{C}}{1 \times 0.512 \text{ }^\circ\text{C kg/mol}} = 0.500 \text{ mol/kg}$$
$$n = 0.500 \text{ mol/kg} \times 0.0500 \text{ kg} = 0.0250 \text{ mol}; M = \frac{3.00 \text{ g}}{0.0250 \text{ mol}} = \mathbf{120. \text{ g/mol}}$$

C. Osmotic pressure

7. Protein.

$$\Pi = \frac{3.80 \text{ torr}}{760 \text{ torr/atm}} = 5.00 \times 10^{-3} \text{ atm}; T = 298.15 \text{ K}$$
$$M_{\text{conc}} = \frac{\Pi}{RT} = \frac{5.00 \times 10^{-3} \text{ atm}}{(0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1})(298.15 \text{ K})} = 2.044 \times 10^{-4} \text{ mol/L}$$
$$n = 2.044 \times 10^{-4} \text{ mol/L} \times 0.1000 \text{ L} = 2.044 \times 10^{-5} \text{ mol}; \text{ molar mass} = \frac{1.00 \text{ g}}{2.044 \times 10^{-5} \text{ mol}} = \mathbf{4.89 \times 10^4 \text{ g/mol}}$$

8. Isotonic saline.

$$T = 37.0 + 273.15 = 310.15 \text{ K}$$
$$M = \frac{\Pi}{iRT} = \frac{7.70 \text{ atm}}{2 \times (0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1})(310.15 \text{ K})} = \mathbf{0.151 \text{ mol/L}}$$

D. Challenge

9. Road salt.

$$m = \frac{\Delta T_f}{iK_f} = \frac{10.0 \text{ }^\circ\text{C}}{2 \times 1.86 \text{ }^\circ\text{C kg/mol}} = 2.688 \text{ mol/kg}, \text{ so } 2.688 \text{ mol in } 1.00 \text{ kg of water}$$
$$\text{mass} = 2.688 \text{ mol} \times 58.44 \text{ g/mol} = \mathbf{157 \text{ g of NaCl}}$$

CaCl_2 gives **three** ions per formula unit ($i = 3$) instead of two, so it lowers the freezing point more per mole, and it can lower the freezing point much further (it is very soluble).